

QUESTION 15:

1. DECISION TREE

Aim

To implement a Decision Tree Classifier for classifying data and predicting the target class.

Algorithm

- 1. Start.**
- 2. Load the dataset.**
- 3. Split the dataset into training and testing sets.**
- 4. Create a Decision Tree classifier.**
- 5. Train the classifier using the training data.**
- 6. Predict the class labels for the test data.**
- 7. Compare the predicted values with the actual values.**
- 8. Calculate the accuracy of the model.**
- 9. Display the predictions and accuracy.**
- 10. Stop.**

Pseudocode

BEGIN

Load Dataset

Split Dataset into Training and Testing Data

Create Decision Tree Model

Train Model using Training Data

Predict Output using Testing Data

Calculate Accuracy

Display Predictions

Display Accuracy

END

OUTPUT :

main.py	   Share	Run	Output
<pre>1 # Simple Decision Tree using if-else 2 3 print("Decision Tree Classification") 4 5 age = int(input("Enter Age: ")) 6 income = int(input("Enter Monthly Income: ")) 7 8 if age < 30: 9 if income >= 50000: 10 print("Prediction: Buy Laptop") 11 else: 12 print("Prediction: Do Not Buy Laptop") 13 else: 14 if income >= 30000: 15 print("Prediction: Buy Laptop") 16 else: 17 print("Prediction: Do Not Buy Laptop")</pre>			Decision Tree Classification Enter Age: 18 Enter Monthly Income: 1000 Prediction: Do Not Buy Laptop === Code Execution Successful ===

Result

The Decision Tree model was successfully implemented. The classifier was trained and tested on the dataset, and it produced predictions with high accuracy.

2. Feed Forward Neural Network (FFNN)

Aim

To implement a Feed Forward Neural Network (FFNN) for classifying data using a Multi-Layer Perceptron (MLP).

Algorithm

- 1. Start.**
- 2. Load the dataset.**
- 3. Split the dataset into training and testing sets.**
- 4. Normalize (scale) the input features.**
- 5. Create a Feed Forward Neural Network model.**
- 6. Train the model using the training data.**
- 7. Predict the output for the test data.**
- 8. Calculate the model accuracy.**
- 9. Display the predicted values and accuracy.**
- 10. Stop.**

Pseudocode

BEGIN

Load Dataset

Split Dataset into Training and Testing Data

Normalize Input Features

Create Feed Forward Neural Network Model

Train Model using Training Data

Predict Output using Testing Data

Calculate Accuracy

Display Predictions

Display Accuracy

OUTPUT:

<pre>1 # Feed Forward Neural Network 2 3 print("Feed Forward Neural Network") 4 5 # Inputs 6 x1 = float(input("Enter Input 1: ")) 7 x2 = float(input("Enter Input 2: ")) 8 9 # Weights and Bias 10 w1 = 0.5 11 w2 = 0.8 12 bias = 0.2 13 14 # Feed Forward Calculation 15 net = (x1 * w1) + (x2 * w2) + bias 16 17 print("Net Input =", net) 18 19 # Activation Function (Step Function) 20 if net >= 1: 21 output = 1 22 else: 23 output = 0 24 25 print("Network Output =", output)</pre>	<pre>Feed Forward Neural Network Enter Input 1: 20 Enter Input 2: 30 Net Input = 34.2 Network Output = 1 === Code Execution Successful ===</pre>
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END

Result

The Feed Forward Neural Network was successfully implemented. The model learned from the training data and classified the test data with high prediction accuracy. These results demonstrate that the FFNN can effectively perform classification tasks.